coherence

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(sec:coherence-interference)= # What is quantum coherence?

0.1 Definition of coherence

In {numref}`sec:coherence-quantization`, we mentioned the difficulty of defining the concept of photon. Similarly, the precise definition of quantum coherence is difficult. Most of textbooks do not touch on this issue. People seems agreeing that

`{prf:definition} Coherence :name: def:coherence Coherence is the ability of a state to generate observable interference.`

How do we mathematically implement the above definition? The state of a quantum system is expressed as a density operator ρ. How do we know if a density operator is capable of generating observable interference? We need to make a certain measurement to see interference. Suppose that we measure an observable $\hat{A}$ whose eigenvalues and the corresponding eigenvectors are expressed as a_j and $|a_j\rangle$. Based on the Born rule, the probability of obtaining a_1 is given by

$$P_1 = \mathrm{tr}\left(|a_1\rangle\langle a_1|\rho\right) = \langle a_1|\rho|a_1\rangle$$

This expression does not expose the presence of coherence. We need to introduce a basis set $|u_j\rangle$. Then, the probability is written as

$$P_1 = \sum_{ij} \langle a_1|u_i\rangle \rho_{ij} \langle u_j|a_1\rangle = \sum_j |\langle a_1|u_j\rangle|^2 \rho_{jj} + \sum_{i\neq j} \langle a_1|u_i\rangle\langle u_j|a_1\rangle \rho_{ij}$$

where the density matrix is defined by $\rho_{ij} = \langle u_i|\rho|u_j\rangle$. The second term in the right hand side is clearly showing interference between $|u_1\rangle$ and $|u_2\rangle$ through the off-diagonal elements $\rho_{i\neq j}$. Therefore, mathematically speaking, if the state has the ability to generate obervable interference or not is determined by the off-diagonal elements of the density operator.

But it depends on the choice of the basis set. If a different basis set is used, the value of the off-diagonal elements change. Furthermore, we can always diagonalize a density matrix, eliminating all off-diagonal elements entirely.

We conclude that coherence is the ability of a state to generate observable interference between *certain basis vectors*. The ability is determined by the off-diagonal elements of the density matrix in *a specified basis set*.

“‘{important} Coherence and basis set :name: adm:coherence-basis

It does not make a sense to discuss coherence without specifying a basis set.

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0.2 Coherence vs superposition

Interference is well-defined phenomena in the classical wave theory. In quantum mechanics, wave functions can generate interference exactly in the same way as the classical waves. It seems that any quantum state can generate interference and thus any quantum state seems coherent. This misleading idea is coming from undergraduate quantum mechanics courses in which all examples are perfectly coherent.

In undergraduate textbooks, the state of a quantum system is mathematically specified by a vector in a Hilbert space. The superposition principle states that any state vector $|\psi\rangle$ can be expressed as a superposition of basis vectors $|u_j\rangle$ as

$$|\psi\rangle = \sum_j c_j |u_j\rangle$$

Let us check if this state is coherent or not based on our definition. Writing the state in terms of density operator

$$\rho = |\psi\rangle\langle\psi| = \sum_{ij} c_i c_j^* |u_i\rangle\langle u_j|$$

The off-diagonal elements are $\rho_{i\neq j} = c_i c_j^*$. All off-diagonal elements vanish if and only if $c_i \neq 0$ and all other coefficients vanish, a state $|\psi\rangle = |u_i\rangle$ cannot generate interference in the current basis set.[1] Otherwise, the state has the ability of generating interference and thus the state is coherent. Does this mean any quantum state is coherent?

It is true that if the state of a system can be written as a superposition, then the system is coherent in that basis. No doubt about that. The question is if all states can be written as a superpoisition of basis vectors. The answer is No. Any state that can be written as a state vector is called a *pure* state and the corresponding density operator is $\rho = |\psi\rangle|\langle\psi|$. There are other states that cannot be expressed with a state vector. Those states are known as a *mixed* state. A mixed state can be also coherent even though it cannot be in a simple superposition state.

In conclusion, any pure state written as a superposition of basis vectors, it is highly coherent. However, not all states are pure. A mixed state can have a certain level of coherency as long as not all off-diagonal elements vanish.

0.3 The origin of decoherence

The fundamental principles of quantum mechanics defines the state of a system as a vector which is inherently coherent in any basis. Then, how is the coherence lost? Actually, we never lose coherence at the fundamental level. Suppose that a system interacts with another system and they form quantum entanglement. The total system is in a superposition state. However, if we look at only the first system, its state always a mixed state due to the entanglement.

[1] If there is only one nonvanishing diagonal element, the state is pure and we have no way to determine coherency from it.

As an example, consider a pair of two-level systems (two qubits). The product basis is

$$|ee\rangle = |e\rangle \otimes |e\rangle, \quad |eg\rangle = |e\rangle \otimes |g\rangle, \quad |ge\rangle = |g\rangle \otimes |e\rangle, \quad |gg\rangle = |g\rangle \otimes |g\rangle$$

(eq:dimer-local-basis)

If the whole system is in an entangled state $|\psi\rangle = (|eg\rangle - |ge\rangle)/\sqrt{2}$, the total system is clearly in a pure state and thus they are coherent. What is the state of the first system? Mathematically it is given by a reduced density operator

$$\rho_1 = \mathrm{tr}_2 |\psi\rangle\langle\psi| = \frac{1}{2}\left(|e\rangle\langle e| + |g\rangle\langle g|\right)$$

where tr_2 is a partial trace over the Hilbert space of the second system.{cite:p}`Audretsch2007` Notice that the off-diagonal element has completely vanished. Hence, the first system has no coherence in the basis of $|e\rangle$ and $|g\rangle$. Actually, this particular state is maximally mixed and the off-diagonal elements vanish in any basis set. The loss of coherence (decoherence) is entirely due to the quantum entanglement between the two systems. The lack of information on the second system causes decoherence in the first system despite that the total system is fully coherent in any basis.

In quantum optics, the electronic system in an emitter is always interacting with radiation fields and they are usually entangled. When we look only at the emitter or only at the radiation field, the state of each system is mixed and coherence is, partially or completely, lost in some basis sets. The loss appears in the off-diagonal elements of the reduced density matrix.

One of the main goal of the present lecture note is to find the reduced density of the emitters using the theory of open quantum system.{cite:p}`Carmichael1993,Breuer2002` and predict the detection of photons using it.

0.4 First-order quantum coherence

Before going to discuss the general coherence functions in {numref}`sec:coherence-quantum`. Here we look at a special case of coherence in the QDs. We again consider a pair of two-level systems. Since the Hilbert space is four dimensional, there are six unique off-diagonal elements in the density matrix, of which $\rho_{eg,gg}$, $\rho_{ge,gg}$, $\rho_{eg,ee}$, and $\rho_{eg,ee}$ are off-diagonal elements representing transition within an individual QD. Only $\rho_{gg,ee}$ and $\rho_{ge,eg}$ are responsible for coherence across the two QDs, of which $\rho_{eg,ge}$ is the most interesting to us since $\rho_{gg,ee}$ envolves two photons in one transition.

How do we measure it? There is a standard way to determine the coherence in optics. Introducing raising (creation) operators for each system, $\sigma_1^+ = |e\rangle\langle g| \otimes I$ and $\sigma_2^+ = I \otimes |e\rangle\langle g|$, and also lowering (annihilation) operators $\sigma_1^- = |g\rangle\langle e| \otimes I$ and $\sigma_2^- = I \otimes |g\rangle\langle e|$ where I is the identity operator.

Noting that $\sigma^+\sigma^-$ acts as a number operator, we compute the population of individual state as correlation function. For example, if we are interested in the emission process in each individual QD, we compute

$$\langle \sigma_1^+ \sigma_1^- \rangle = \mathrm{tr}(\sigma_1^+ \sigma_1^- \rho) = \mathrm{tr}[(|e\rangle\langle e| \otimes I)\rho] = \langle e|\rho_1|e\rangle$$

which is the probability of the system 1 in the excited state $|e\rangle$. Similarly, we can calculate the probability of the system 1 in the ground state $|g\rangle$ by

$$\langle\sigma_1^-\sigma_1^+\rangle = \langle g|\rho_1|g\rangle.$$

Notice the order of raising and lowering operators is reversed.

The above calculation is useful if the two QDs operate independently. If they emit a photon jointly, the story is completely different as we see in suparradiance. How do we know they work together? The following cross-correlation tells coherence between the systems.

$$\langle\sigma_1^+\sigma_2^-\rangle = \mathrm{tr}(\sigma_1^+\sigma_2^-\rho) = \rho_{ge,eg}.$$

If this cross-correlation does not vanish, interference between the systems takes place (in the product basis set). Throughout the lecture note, we will see this kind of cross correlations because we are interested in *collective* photoemission.

Many of us have a strong preoccupied conception of photon emitted from individual QDs and tend to use the product basis. However, from theoretical perspectives, a basis set that is consistent with the entanglement between the emitter and the radiation field. For example, the following basis set takes into account the symmetry ($D_{\infty h}$) of two QDs,

$$|u_1\rangle = |ee\rangle, \quad |u_2\rangle = \frac{1}{\sqrt{2}}\left(e^{-i\phi}|eg\rangle + e^{i\phi}|ge\rangle\right), \quad |u_3\rangle = \frac{1}{\sqrt{2}}\left(e^{-i\phi}|eg\rangle - e^{i\phi}|ge\rangle\right), \quad |u_4\rangle = |gg\rangle,$$

To make the vectors invariant under the symmetry operation, ϕ changes its sign when the QDs are exchanged. Whereas $|u_1\rangle$, $|u_2\rangle$, and $|u_4\rangle$ are symmetric with respect to the exchange of QDs, u_3 is anti-symmetric. When $\phi_j = 0$ it is known as the Dicke basis.

The corresponding raising operators are

$$\begin{aligned}
\eta_1^+ &= |u_2\rangle\langle u_4| + |u_1\rangle\langle u_2| = \frac{1}{\sqrt{2}}\left(e^{-i\phi}\sigma_1^+ + e^{i\phi}\sigma_2^+\right) \\
\eta_2^+ &= |u_3\rangle\langle u_4| + |u_1\rangle\langle u_3| = \frac{1}{\sqrt{2}}\left(e^{-i\phi}\sigma_1^+ - e^{i\phi}\sigma_2^+\right)
\end{aligned}$$

and the lowering operators are their conjugate $\eta_j^- = (\eta_i^+)^\dagger$. The raising operator η_1^+ raises the state from $|u_4\rangle$ to $|u_2\rangle$ and also from $|u_2\rangle$ to $|u_1\rangle$. We shall call it channel 1. Similarly, in channel 2, η_2^+ raises the state from from $|u_4\rangle$ to $|u_3\rangle$ and also from $|u_3\rangle$ to $|u_1\rangle$. Both channels involve two QDs simultaneously and thus represents collective transitions.

Now, the intensity of emission from channel 1 is proportional to the population of $|u_1\rangle$ and $|u_2\rangle$. Noting that $\eta_1^+\eta_1^-$ is the corresponding number operator, we can calculate the popoulation by $\langle\eta_1^+\eta_1^-\rangle = \rho_{11} + \rho_{22}$. Similarly for channel 2, $\langle\eta_2^+\eta_2^-\rangle = \rho_{11} + \rho_{33}$. Theoretically it is very simple.

If the product basis is preferred, we have

$$\begin{aligned}
\langle\eta_1^+\eta_1^-\rangle &= \frac{1}{2}\left(\langle\sigma_1^+\sigma_1^-\rangle + \langle\sigma_2^+\sigma_2^-\rangle + e^{-2i\phi}\langle\sigma_1^+\sigma_2^-\rangle + e^{+2i\phi}\langle\sigma_2^+\sigma_1^-\rangle\right) \\
\langle\eta_2^+\eta_2^-\rangle &= \frac{1}{2}\left(\langle\sigma_1^+\sigma_1^-\rangle + \langle\sigma_2^+\sigma_2^-\rangle - e^{-2i\phi}\langle\sigma_1^+\sigma_2^-\rangle - e^{+2i\phi}\langle\sigma_2^+\sigma_1^-\rangle\right)
\end{aligned}$$

(eq:eta-population)

Clearly, the off-diagonal element $\langle\sigma_1^+\sigma_2^-\rangle = \rho_{eg,ge}$ is envolved and the difference between the channels is the sign of interference. If the symmetry-adapted basis is used, the interference is automatically taken into account. On the other hand, the product basis is commonly used in the numerical approach.

Equation {eq}`eq:eta-population` is known as the first-order correlation.

0.5 Second-order quantum coherence

The second-order coherence measures the "interference" of intensity rather than amplitude. That means we measure the square of intensity.

$$\langle(\eta_1^+\eta_1^-)^2)\rangle = \sum_{ij}\sum_{k\ell}(c_i^*c_j)(c_k^*c_\ell)\langle\sigma_i^+\sigma_k^+\sigma_\ell^-\sigma_j^-\rangle$$

where $c_1 = e^{-i\phi}$ and $c_2 = e^{i\phi}$. This is called the second-order coherence. Again, in the symmetry adopted basis, we just calculate a single correlation but in the product basis, we need to sum up many terms. Again, it has theoretical advantage but the product basis is commonly used in numerical calculations. We will see the second order correlation through out this lecture note.

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